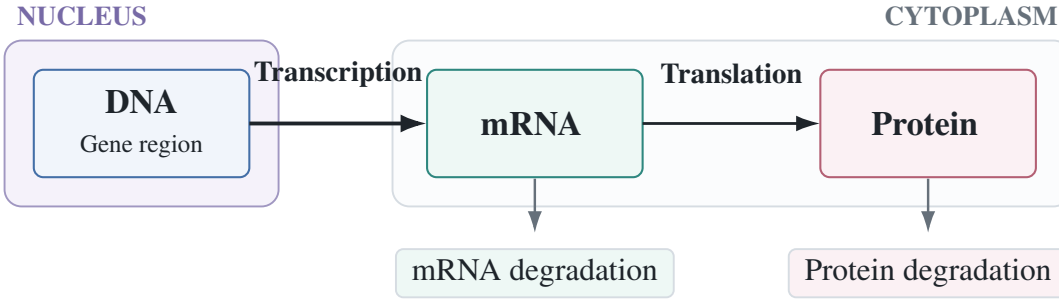


Gene Expression Model

$$\mathrm{S}_i \xrightarrow{\widetilde{a}_{i,j}} \mathrm{S}_j \quad (i \neq j,\ 1 \leq i, j \leq N)$$

$$\mathrm{S}_i \xrightarrow{\widetilde{b}_{i,j}} \mathrm{S}_j + \mathrm{mRNA} \quad (1 \leq i, j \leq N)$$

$$\mathrm{mRNA} \xrightarrow{u} \emptyset$$

$$\mathrm{mRNA} \xrightarrow{v} \mathrm{mRNA} + \mathrm{Protein}$$

$$\mathrm{Protein} \xrightarrow{\delta} \emptyset$$

Burst Approximation

Gene Expression Model under the Burst Approximation

$$\mathrm{S}_i \xrightarrow{a_{i,j}} \mathrm{S}_j \quad (i \neq j,\ 1 \leq i, j \leq N)$$

$$\mathrm{S}_i \xrightarrow{b_{i,j}^{[r]}} \mathrm{S}_j + r \cdot \mathrm{Protein} \quad (1 \leq i, j \leq N,\ r \geq 1)$$

$$\mathrm{Protein} \xrightarrow{\delta} \emptyset$$